

International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

Case Studies: Investigating Harassment Email Traffic With Wireshark

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EXECUTIVE SUMMARY

This investigation examined a network traffic capture associated with the investigation of harassment messages received by Lily Tuckrige, a Chemistry 109 instructor at Nitroba State University. The objective was to determine whether the captured network traffic provided evidence linking the later harassment activity to a device, an identifiable user account, and a student enrolled in Chemistry 109.

The investigation was conducted using Wireshark and supporting command-line utilities on a Kali Linux forensic workstation. The supplied Nitroba PCAP was acquired into a dedicated case directory and its SHA-256 hash was calculated and compared with the published hash for the case evidence. The calculated value, 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb, matched the published value, providing an integrity check for the acquired capture.

Packet analysis identified Frame 83601 as the relevant HTTP POST /secure/submit request to www.willselfdestruct.com. The request originated from client IP address 192.168.15.4 and Ethernet MAC address 00:17:f2:e2:c0:ce. The decoded HTTP form data contained the recipient lilytuckrige@yahoo.com, the subject you can't find us, and the harassment message beginning and you can't hide from us. followed by Stop teaching. and Start running.

Further analysis of traffic originating from the same MAC address identified Frame 79187, which contained the HTTP cookie gmailchat=jcoachj@gmail.com/475090. This provided

browser/session evidence associating the observed device with the Gmail account jcoachj@gmail.com. The supplied Chemistry 109 class list contains a student named Johnny Coach, providing an additional association between the identified account and a student taught by Lily Tuckrige.

The evidence therefore provides a strong association between the harassment request, the identified network device, the Gmail account, and a Chemistry 109 student. However, the evidence does not conclusively establish that Johnny Coach personally operated the device at the precise time of the harassment request. The case scenario indicates that the wireless network was open and accessible to multiple people, and a browser cookie establishes an account/session association rather than definitive proof of the physical user. The conclusion is therefore limited to the level supported by the available network and browser evidence.

PURPOSE OF THE ASSIGNMENT

The purpose of this assignment is to conduct a computer forensic examination of captured network traffic associated with a harassment investigation. The examination focuses on identifying the client system that communicated with the relevant web service, correlating the network traffic with the harassment message, identifying the network device involved, and determining whether available browser and account evidence can associate the device with an individual enrolled in Chemistry 109. The investigation also demonstrates the application of forensic evidence acquisition, integrity verification, packet analysis, evidence correlation, timeline analysis, and cautious attribution.

OBJECTIVES OF THE ASSIGNMENT

The objectives of this investigation are to:

1. Acquire and preserve the supplied network traffic capture in a structured forensic working environment.
2. Verify the integrity and technical characteristics of the acquired PCAP.
3. Identify the client system that communicated with the willselfdestruct.com web service.
4. Identify the network traffic containing the harassment message submitted to Lily Tuckrige's email address.

5. Identify the Ethernet MAC address associated with the relevant client traffic.
6. Examine browser and HTTP cookie information for evidence linking the device to an identifiable account.
7. Compare the identified account with the supplied Chemistry 109 student roster.
8. Establish the timestamp of the relevant network activity using the PCAP's recorded timestamps.
9. Assess the strength and limitations of the evidence for attributing the harassment activity to an individual.
10. Document the investigation in a traceable manner using packet numbers, timestamps, display filters, screenshots, and an evidence log.

METHODOLOGY OF THE ASSIGNMENT

The investigation followed a structured network forensic examination process. First, a dedicated case directory was established on the Kali Linux workstation to separate the working PCAP, derived evidence, screenshots, and investigation notes. The supplied Nitroba network traffic capture was then acquired into the working directory and its file size and cryptographic hash were verified. The PCAP metadata was subsequently examined using command-line utilities before packet-level analysis was conducted in Wireshark.

Wireshark display filters were used to progressively narrow the captured traffic to communications involving www.willselfdestruct.com. HTTP GET and POST requests were examined to identify the client system and the relevant submission request. The HTTP form data in the identified POST request was inspected to establish the recipient, subject, and message content. Ethernet and IP information were then correlated to identify the MAC address and client IP associated with the request.

Additional traffic originating from the identified MAC address was examined for HTTP cookie information. A Gmail cookie containing `gmailchat=jcoachj@gmail.com/475090` was identified and used as browser/session evidence associating the device with the Gmail account. The account identifier was then compared with the supplied Chemistry 109 class list. Finally, the packet timestamp for the relevant harassment request was

examined using Wireshark's date-and-time display and Frame details. Findings were documented using packet numbers, timestamps, display filters, screenshots, and evidence-log entries. The final attribution assessment was limited to conclusions supported by the captured evidence and the stated limitations of the shared wireless environment.

TOOLS AND RESOURCES USED

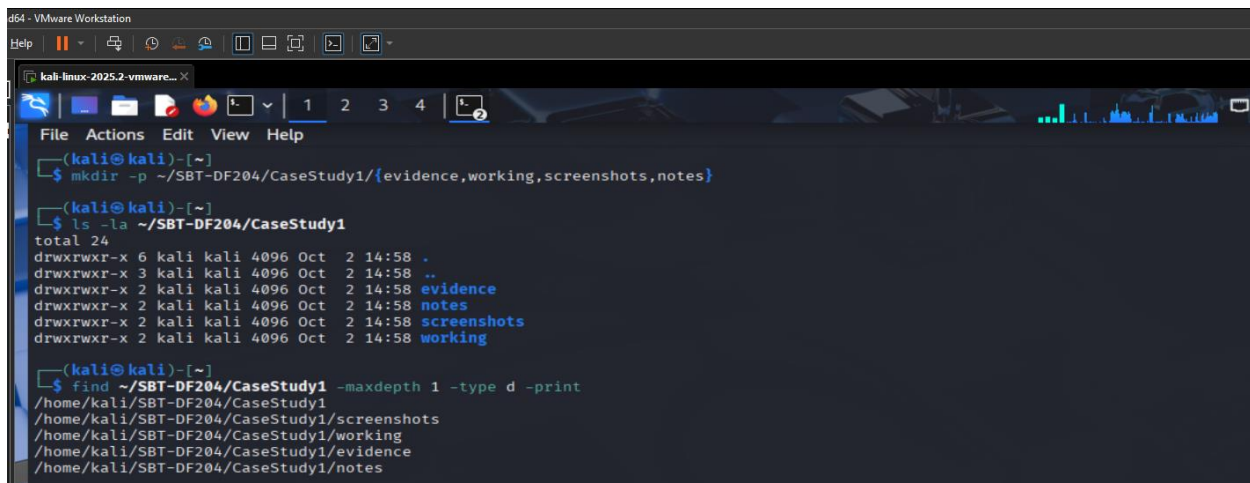
The following tools and resources were used during the investigation:

Tool / Resource	Purpose / Use in the Investigation
Kali Linux	Provided the forensic examination workstation and working environment.
Wireshark	Used for packet capture analysis, display filtering, HTTP inspection, MAC/IP correlation, cookie analysis, and timestamp examination.
sha256sum	Used to calculate the SHA-256 cryptographic hash of the acquired PCAP for integrity verification.
capinfos	Used to verify PCAP metadata, including packet count, file size, capture duration, encapsulation type, timestamp precision, and capture order.
file	Used to independently identify and verify the format and characteristics of the acquired PCAP file.
wget	Used to acquire the Nitroba PCAP from the specified Digital Corpora source.
find	Used to verify the structure of the dedicated forensic case directories.
ls -lh	Used to confirm the presence and reported size of the acquired PCAP file.
Nitroba PCAP (nitroba.pcap)	Served as the primary network evidence examined during the investigation.

Tool / Resource	Purpose / Use in the Investigation
Chemistry 109 Class List	Used as supporting case material to determine whether the individual associated with the identified Gmail account was enrolled in Lily Tuckrige's Chemistry 109 class.

Establishment of the Forensic Working Environment

The forensic investigation was initiated by creating a dedicated case directory on the Kali Linux workstation. A structured directory hierarchy was established under `/home/kali/SBT-DF204/CaseStudy1/` to maintain separation between the different types of investigation material. Four subdirectories were created: evidence, working, screenshots, and notes. The working directory was designated for the PCAP file used during the examination, while the screenshots directory was reserved for visual evidence captured during the investigation. The notes directory was created to preserve investigation notes and supporting command outputs, while the evidence directory was reserved for other relevant case materials. This structure was established before commencing the technical analysis in order to maintain an organised forensic workspace and to make the subsequent examination and reporting process traceable.



```

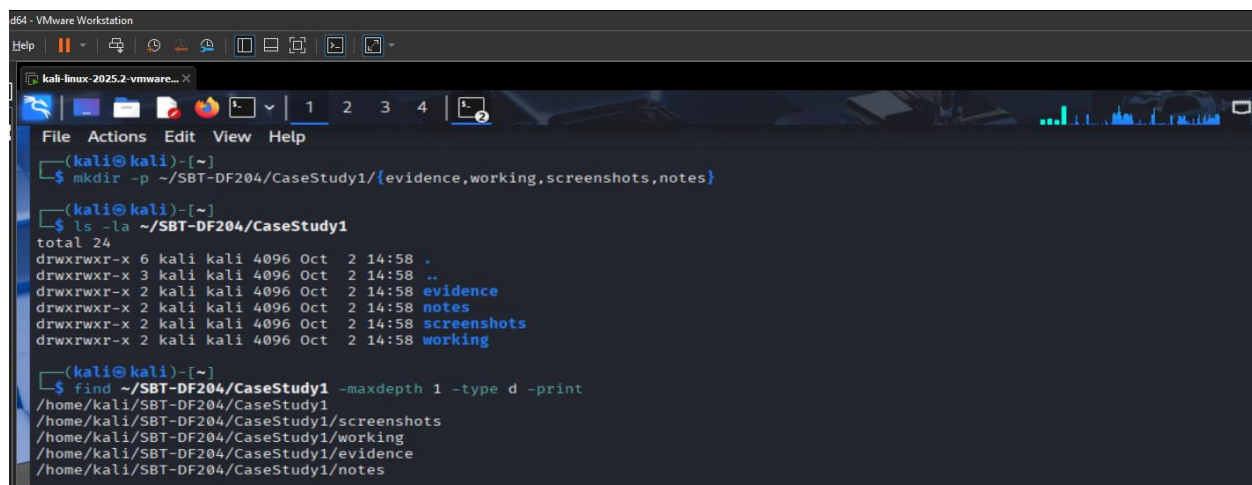
kali-linux-2025.2-vmware...
File Actions Edit View Help
(kali@kali)-[~]
$ mkdir -p ~/SBT-DF204/CaseStudy1/{evidence,working,screenshots,notes}
(kali@kali)-[~]
$ ls -la ~/SBT-DF204/CaseStudy1
total 24
drwxrwxr-x 6 kali kali 4096 Oct  2 14:58 .
drwxrwxr-x 3 kali kali 4096 Oct  2 14:58 ..
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 evidence
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 notes
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 screenshots
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 working
(kali@kali)-[~]
$ find ~/SBT-DF204/CaseStudy1 -maxdepth 1 -type d -print
/home/kali/SBT-DF204/CaseStudy1
/home/kali/SBT-DF204/CaseStudy1/screenshots
/home/kali/SBT-DF204/CaseStudy1/working
/home/kali/SBT-DF204/CaseStudy1/evidence
/home/kali/SBT-DF204/CaseStudy1/notes

```

Figure 1: Creation and verification of the dedicated forensic case directory structure.

Verification of the Case Directory Structure

Following creation of the case directories, the directory structure was independently verified using the find command. The command confirmed that the case directory and its four designated subdirectories existed at the expected location: /home/kali/SBT-DF204/CaseStudy1/evidence, /home/kali/SBT-DF204/CaseStudy1/working, /home/kali/SBT-DF204/CaseStudy1/screenshots, and /home/kali/SBT-DF204/CaseStudy1/notes. This verification established that the intended forensic workspace was available before acquisition of the network traffic capture. The use of a dedicated case directory also ensured that the investigation materials were kept separate from unrelated files on the workstation.



```
d64 - VMware Workstation
Help  [Icons]

kali-linux-2025.2-vmware...
File Actions Edit View Help

(kali@kali)-[~]
$ mkdir -p ~/SBT-DF204/CaseStudy1/{evidence,working,screenshots,notes}

(kali@kali)-[~]
$ ls -la ~/SBT-DF204/CaseStudy1
total 24
drwxrwxr-x 6 kali kali 4096 Oct  2 14:58 .
drwxrwxr-x 3 kali kali 4096 Oct  2 14:58 ..
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 evidence
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 notes
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 screenshots
drwxrwxr-x 2 kali kali 4096 Oct  2 14:58 working

(kali@kali)-[~]
$ find ~/SBT-DF204/CaseStudy1 -maxdepth 1 -type d -print
/home/kali/SBT-DF204/CaseStudy1
/home/kali/SBT-DF204/CaseStudy1/screenshots
/home/kali/SBT-DF204/CaseStudy1/working
/home/kali/SBT-DF204/CaseStudy1/evidence
/home/kali/SBT-DF204/CaseStudy1/notes
```

Figure 2: Verification of the forensic case directory and investigation subdirectories.

Acquisition of the Network Traffic Capture

The network traffic capture required for the investigation was acquired from the Digital Corpora source specified in the assessment brief. The file was downloaded directly into the designated working directory using the wget command over HTTPS.

The source specified for the acquisition was <https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap>, and the resulting file was saved as nitroba.pcap. The download process returned an HTTP 200 OK response, indicating that the requested resource was successfully retrieved from the specified source. The server reported a content length of 56,180,821 bytes, and the completed download confirmed that the same number of bytes had been

saved to the working directory. The assessment requires the investigator to record the original filename, source, file size, download date and integrity hash for the traffic capture.

```
(kali@kali)~$ wget -O ~/SBT-DF204/CaseStudy1/working/nitroba.pcap \
https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-02 15:00:38-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.211.137, 52.92.243.201, 52.92.224.89, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.211.137|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: '/home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap'

/home/kali/SBT-DF204/Cas 90%[=====] 48.46M --KB/s eta 5m 33s
```

Figure 3: Acquisition of the Nitroba network traffic capture from the specified Digital Corpora source.

Confirmation of the Acquired PCAP File

After the download was completed, the presence and size of the acquired PCAP were verified using the `ls -lh` command. The resulting output confirmed that the file was located at `/home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap` and that the operating system reported its size as approximately 54 MB. The exact number of bytes reported during the download was 56,180,821 bytes. The difference between the human-readable 54M representation displayed by `ls` and the exact byte count is a result of the way file sizes are represented in command-line output and does not indicate a discrepancy in the downloaded file. At this stage, the PCAP had been successfully acquired and located in the designated working directory for subsequent forensic examination.

```
(kali@kali)~$ wget -O ~/SBT-DF204/CaseStudy1/working/nitroba.pcap \
https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-02 15:00:38-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.211.137, 52.92.243.201, 52.92.224.89, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.211.137|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: '/home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap'

DF204/CaseStudy1/working/nitroba.pcap 90%[=====] 48.48M --KB/s eta 5m 34s
/home/kali/SBT-DF204/CaseStudy1/workin 91%[=====] 49.14M --KB/s eta 5m 50s
/home/kali/SBT-DF204/CaseStudy1/workin 100%[=====] 53.58M 15.8KB/s in 77m 12s

2026-10-02 16:17:54 (11.8 KB/s) - '/home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap' saved [56180821/56180821]

(kali@kali)~$ ls -lh ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Nov 21 2020 /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap

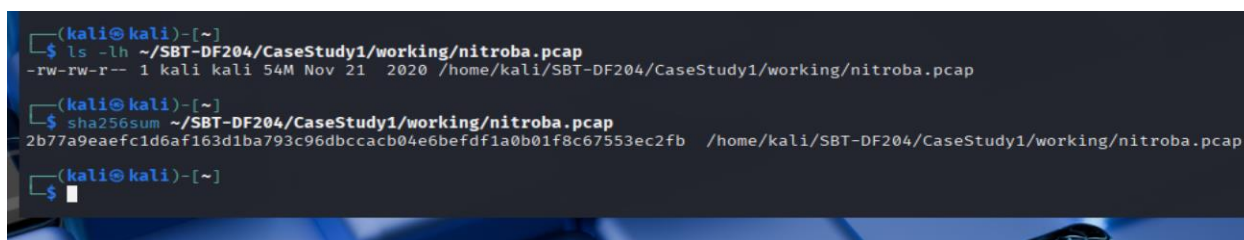
(kali@kali)~$
```

Figure 4: Verification of the acquired PCAP filename, location and file size.

SHA-256 Hash Calculation

Before beginning packet-level examination, a SHA-256 cryptographic hash was calculated for the acquired PCAP using the `sha256sum` utility. The calculated hash was `2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`.

The assessment material provides a published SHA-256 value for the Nitroba PCAP, which is the same value: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb. The exact match between the locally calculated hash and the published hash provides an integrity check indicating that the acquired file corresponds to the PCAP identified by the scenario materials. The hash value will therefore be retained as part of the evidence record and used to identify the specific PCAP examined during this investigation.

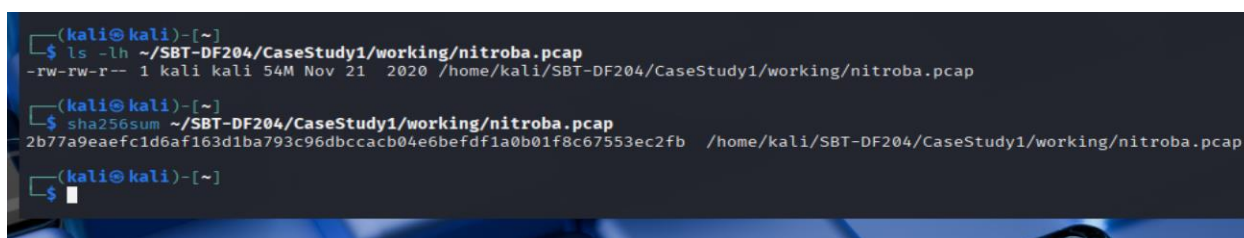


```
(kali@kali)-[~]
└─$ ls -lh ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Nov 21 2020 /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
(kali@kali)-[~]
└─$ sha256sum ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
(kali@kali)-[~]
└─$
```

Figure 5: SHA-256 hash calculated for the acquired Nitroba PCAP.

Evidence Integrity Assessment

The acquisition and initial integrity verification established the foundation for the subsequent forensic examination. The PCAP was obtained from the source specified in the assessment material, stored within the dedicated case working directory, and subjected to cryptographic hash verification. The calculated SHA-256 value exactly matched the published SHA-256 value provided for the scenario PCAP. This provides a documented integrity checkpoint before the packet capture is subjected to further analysis. The subsequent investigation will therefore use the verified nitroba.pcap file as the working evidence source, with material findings documented using packet numbers, timestamps, display filters and supporting screenshots as required by the assessment brief.



```
(kali@kali)-[~]
└─$ ls -lh ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Nov 21 2020 /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
(kali@kali)-[~]
└─$ sha256sum ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
(kali@kali)-[~]
└─$
```

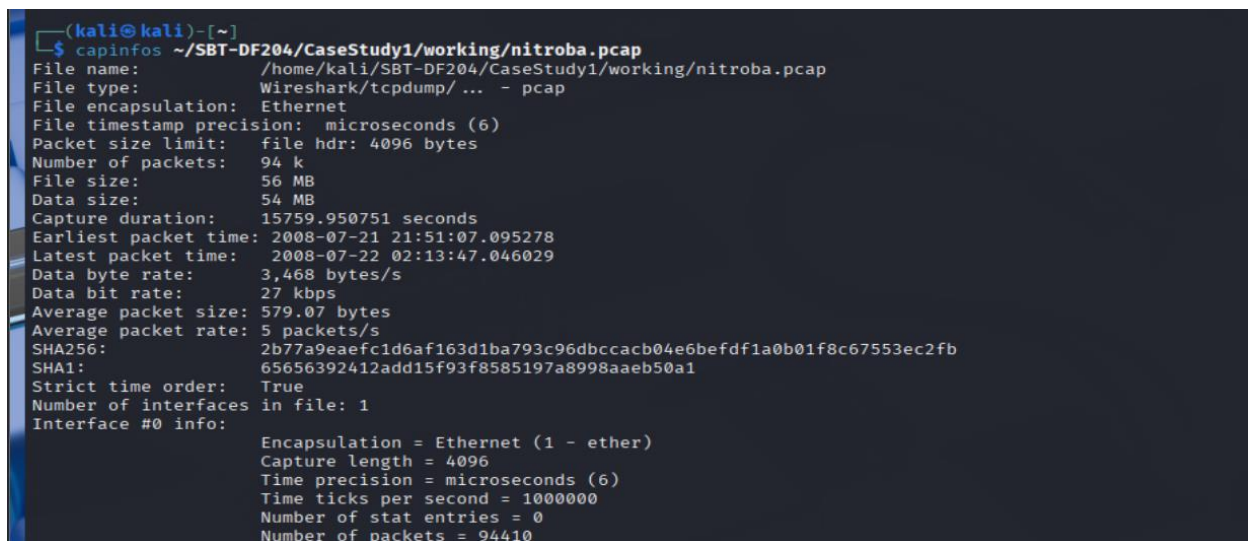
Figure 6: Summary of the verified forensic working evidence.

Evidence Log Entry

The first evidence-log entry was created to document the acquisition and integrity verification of the PCAP. Evidence item E01 represents the nitroba.pcap file and records its successful acquisition and hash verification. The evidence log will be expanded throughout the investigation as individual packets, TCP streams, MAC addresses, HTTP requests, browser or cookie information, and other material findings are identified. This approach ensures that each significant finding can be traced to its corresponding packet or evidence item and supporting screenshot.

PCAP Metadata Verification

The acquired network traffic capture was examined using capinfos to verify its technical characteristics before analysis. The file was confirmed as a Wireshark/tcpdump PCAP with Ethernet encapsulation and microsecond timestamp precision. The capture contains 94,410 packets, has a file size of approximately 56 MB, and contains approximately 54 MB of captured data. The capture begins at 2008-07-21 21:51:07.095278 and ends at 2008-07-22 02:13:47.046029, giving a capture duration of approximately 4 hours, 22 minutes, and 40 seconds. The capinfos output also confirmed that the capture is in strict time order and contains a single captured interface.



```
(kali@kali)-[~]
$ capinfos ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
File name: /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410
```

The SHA-256 value reported by capinfos was 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb, which is identical to the published SHA-256 value provided in the case study. This provides an

additional verification that the file being analyzed is the expected PCAP and has not changed since acquisition.

```
(kali@kali)~$ capinfos ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
File name: /home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 4096
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 94410

(kali@kali)~$ file ~/SBT-DF204/CaseStudy1/working/nitroba.pcap
/home/kali/SBT-DF204/CaseStudy1/working/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
```

The file utility independently identified the evidence as a PCAP capture file, version 2.4, Ethernet encapsulation, with microsecond timestamps and a 4096-byte capture length. This corroborates the file format information reported by capinfos.

Identification of the Client System Communicating With the Web Service

A Wireshark capture was conducted to identify all potential client IP, service IP, packet number, timestamp, and relevant HTTP request, amongst others.

The screenshot displays the Wireshark network protocol analyzer interface. The main window shows a packet capture of 'nitroba.pcap'. The packet list on the left contains 15 packets, with the first packet (No. 1) selected. The packet details pane on the right shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol headers. The packet bytes pane at the bottom shows the raw data of the selected packet. The status bar at the bottom indicates 'Packets: 94410' and 'Profile: Default'.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.64	74.125.19.83	TCP	70	42760 → 80 [FIN, ACK] Seq=7080
2	0.008450	74.125.19.83	192.168.1.64	TCP	70	80 → 42760 [FIN, ACK] Seq=7080
3	0.019619	192.168.1.64	74.125.19.19	HTTP	1421	GET /mail/?logon&hl=en
4	0.044170	74.125.19.19	192.168.1.64	TCP	70	80 → 35011 [ACK] Seq=135
5	0.224402	74.125.19.19	192.168.1.64	HTTP	1284	HTTP/1.1 302 Moved Temporarily
6	0.226712	192.168.1.64	74.125.19.19	TCP	70	35011 → 80 [ACK] Seq=135
7	0.231239	192.168.1.64	74.125.19.19	TCP	70	35011 → 80 [FIN, ACK] Seq=135
8	0.240276	74.125.19.19	192.168.1.64	TCP	70	80 → 35011 [FIN, ACK] Seq=135
9	0.280893	192.168.1.64	192.168.1.254	DNS	78	Standard query 0x61ce A
10	0.283114	192.168.1.64	192.168.1.254	DNS	78	Standard query 0x7362 A
11	0.294021	192.168.1.254	192.168.1.64	DNS	386	Standard query response
12	0.295200	192.168.1.254	192.168.1.64	DNS	78	Standard query response
13	0.308778	192.168.1.64	192.168.1.254	DNS	80	Standard query 0xf689 A
14	0.321240	192.168.1.254	192.168.1.64	DNS	80	Standard query response
15	0.328385	192.168.1.64	74.125.19.103	TCP	82	39153 → 443 [SYN] Seq=

A Wireshark display filter was applied to identify HTTP GET requests directed to the web service specified in the case study: `http.request.method == "GET" && http.host == "www.willselfdestruct.com"`. The filter returned nine HTTP requests. Examination of the resulting traffic showed that the client system involved in the communication used the private IP address 192.168.15.4, while the destination web service resolved in the captured traffic to 69.25.94.22.

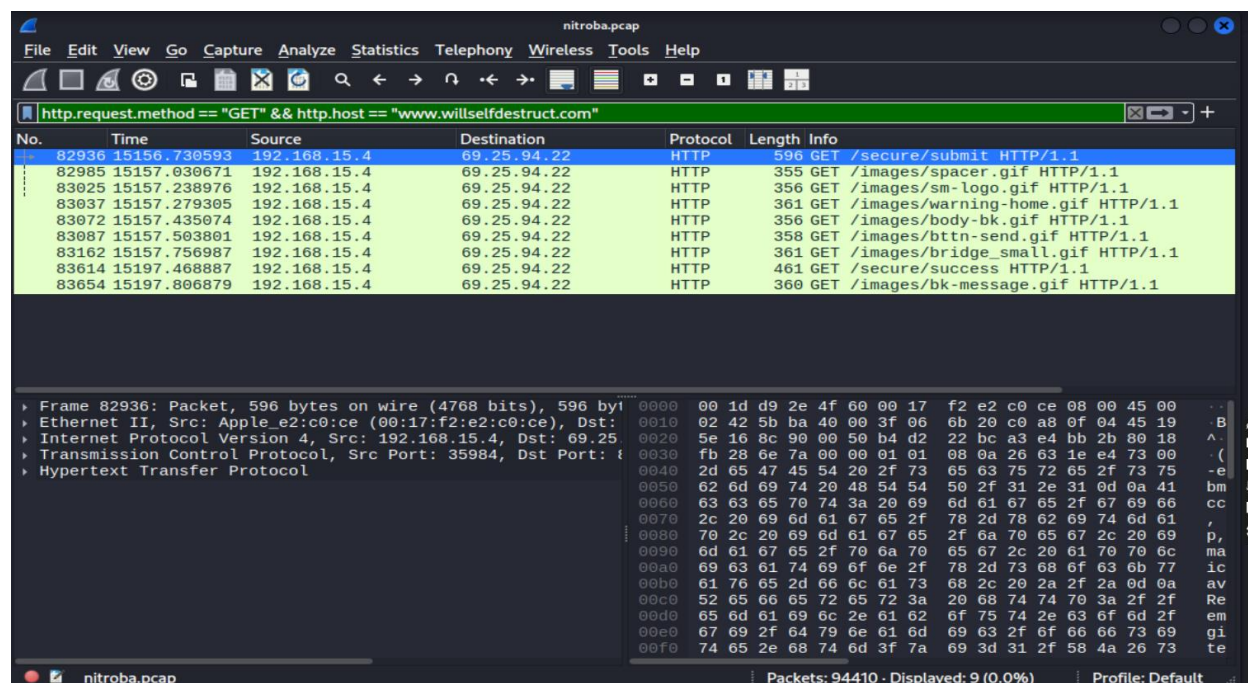


Figure 7: Wireshark identification of client IP 192.168.15.4 communicating with web service 69.25.94.22, including the GET /secure/submit HTTP/1.1.

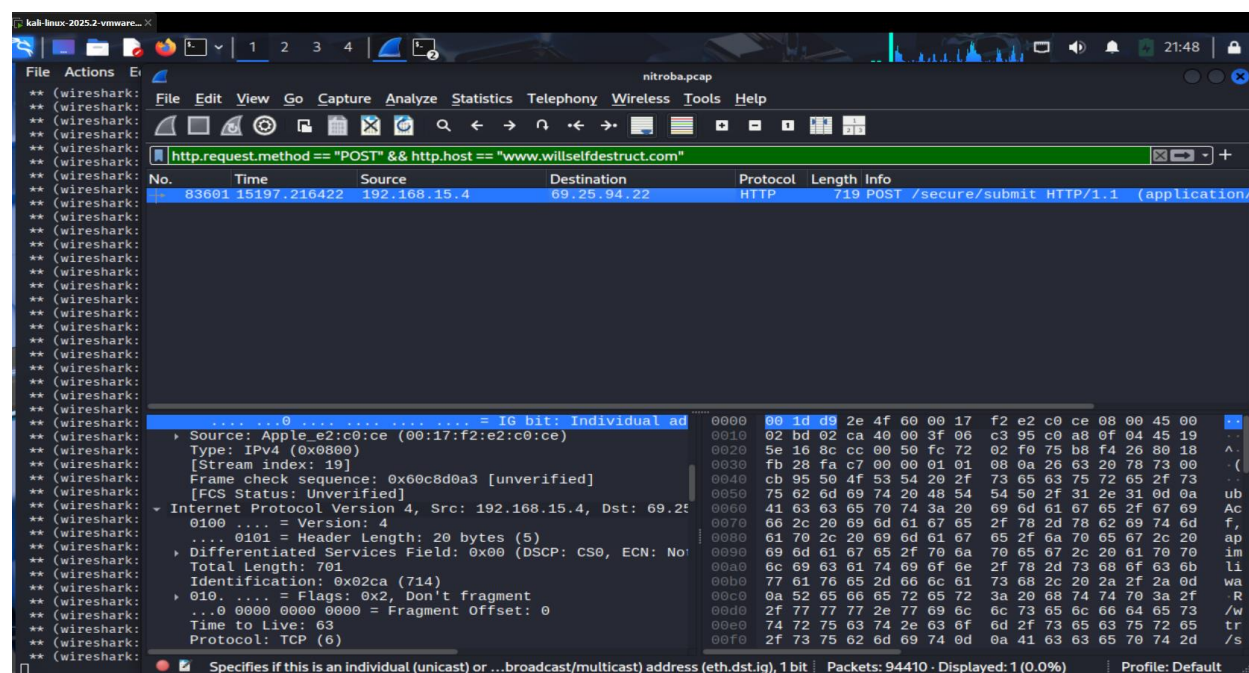
From the screenshot above, we can establish that the filtered traffic showed:

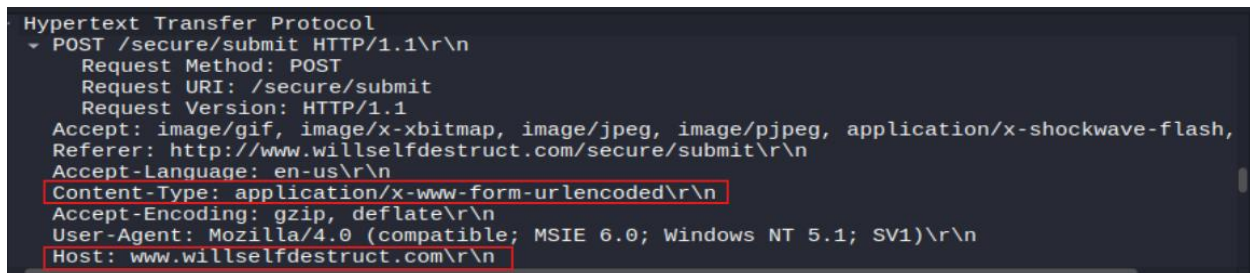
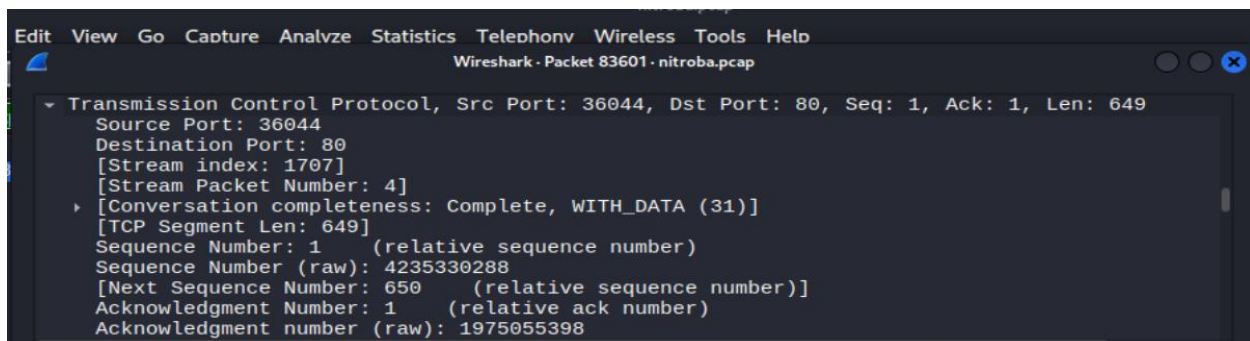
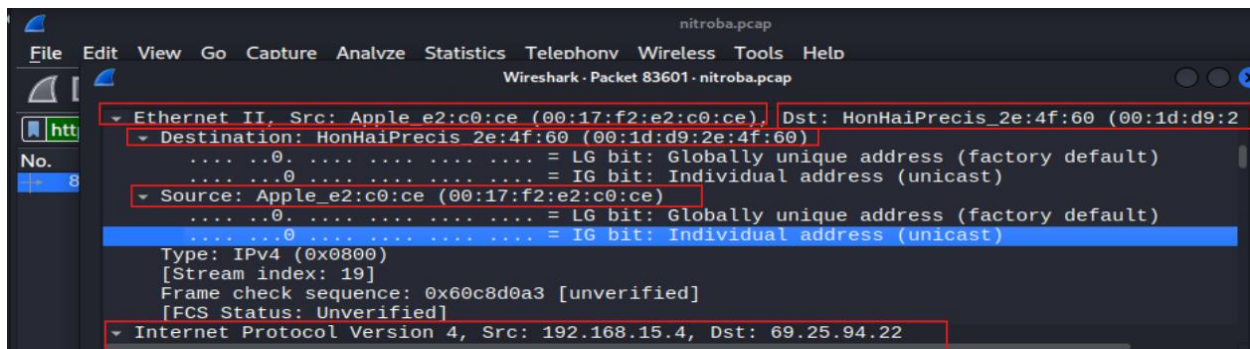
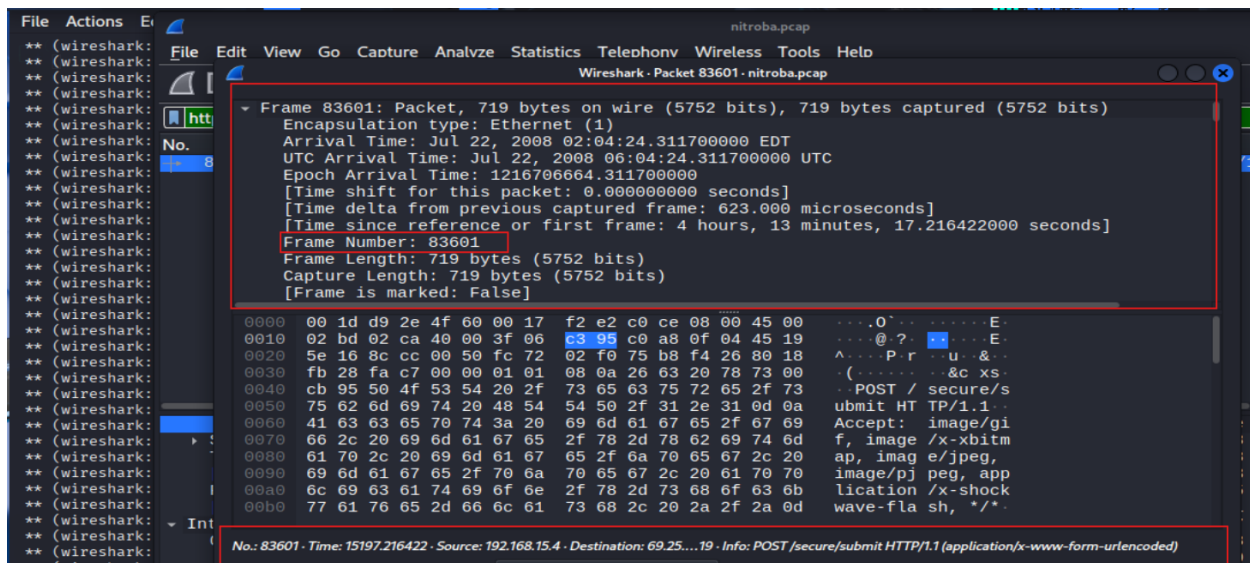
Client IP	192.168.15.4
Web service IP	69.25.94.22
Relevant packet	82936
Timestamp shown by Wireshark	15156.730593 seconds into the capture
Request	GET /secure/submit HTTP/1.1
Protocol	HTTP

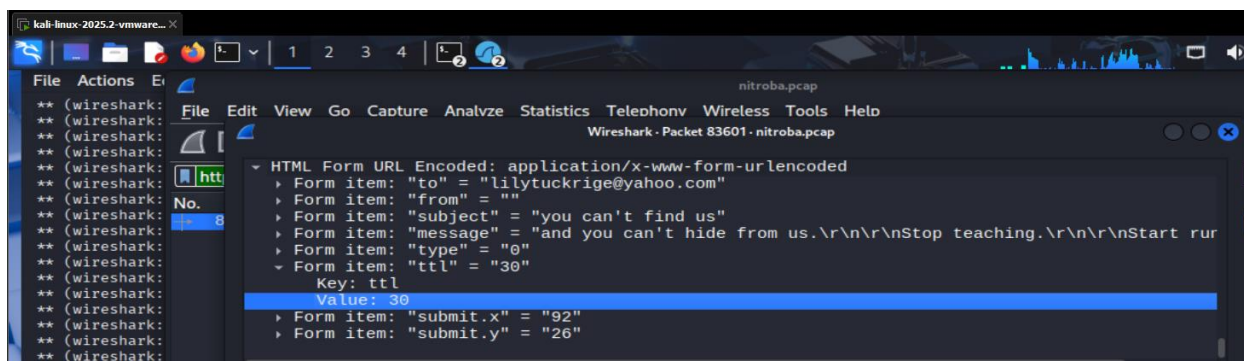
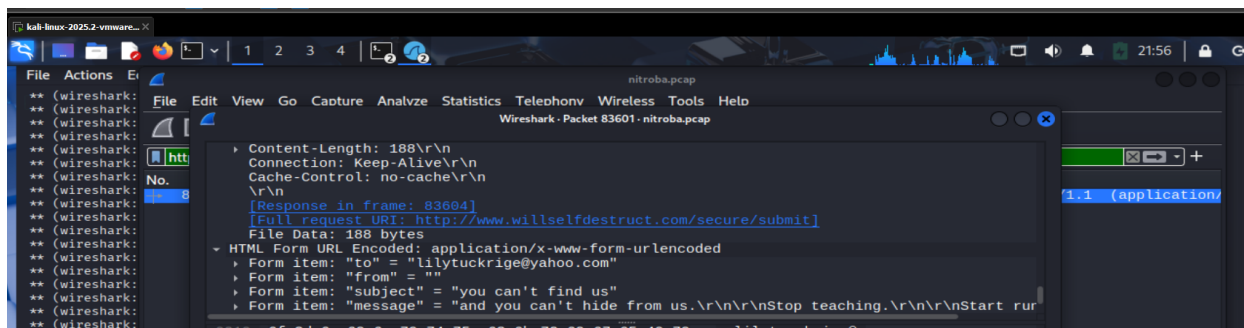
The same client 192.168.15.4 subsequently requests /secure/success and several page images.

The Packet 82936 identified as a particularly relevant request because it contains GET /secure/submit HTTP/1.1, indicating communication with the application's submission endpoint. The packet details further show that the traffic was carried over Ethernet and that the source MAC address associated with the frame begins with 00:17:f2:e2:.... The subsequent filtered requests from the same client include requests for /secure/success and supporting page resources, demonstrating continued interaction between the client and the web service. At this stage, the evidence establishes that 192.168.15.4 was the client IP communicating with the specified web service during the captured session; it does not, by itself, establish which individual was operating the device.

Correlation of the Client System With the Harassment Message







Further examination was performed using the display filter `http.request.method == "POST" && http.host == "www.willselfdestruct.com"`. The filter returned a single relevant HTTP POST request, frame 83601, originating from client IP address 192.168.15.4 and destined for web service IP address 69.25.94.22. The request was sent to `http://www.willselfdestruct.com/secure/submit` over HTTP and used the `application/x-www-form-urlencoded` content type. Wireshark decoded the submitted form data and identified the recipient as lilytuckrige@yahoo.com, the subject as “you can't find us”, and the message beginning “and you can't hide from us.” followed by “Stop teaching.” and “Start run...”. The packet therefore provides direct network-level evidence that the client at 192.168.15.4 submitted a message through the identified web service to the email address associated with the complainant in the case scenario. The packet also records the Ethernet source address as 00:17:f2:e2:c0, providing a link between the IP-layer client and the physical network interface observed in the capture.

What packet 83601 establishes from the output including the following findings:

Frame	83601
Time	15197.216422 seconds into the capture

Source IP	192.168.15.4
Destination IP	69.25.94.22
Source MAC	00:17:f2:e2:c0:ce
Destination MAC	00:1d:d9:2e:4f:60
Source port	36044
Destination port	80
Request	POST /secure/submit HTTP/1.1
Full URI	http://www.willselfdestruct.com/secure/submit
Content type	application/x-www-form-urlencoded
Form data	188 bytes
Recipient	lilytuckrige@yahoo.com
Subject	you can't find us
Message	and you can't hide from us. \r\n\r\nStop teaching. \r\n\r\nStart running

Identification of the Device That Made the Relevant Request

The device associated with the relevant harassment request was identified using its Ethernet Media Access Control (MAC) address. Wireshark filtering with `eth.src == 00:17:f2:e2:c0:ce && ip.dst == 69.25.94.22` identified traffic originating from MAC address 00:17:f2:e2:c0:ce and using client IP address 192.168.15.4 to communicate with the web service at 69.25.94.22. In particular, Frame 83601 shows the same source MAC address and client IP address associated with the HTTP POST /secure/submit request containing the harassment message. This establishes a network-level association between the captured request and the device/network interface identified by MAC address 00:17:f2:e2:c0:ce. However, the MAC address and IP address alone cannot establish which individual was operating the device at the time of the request. This limitation is significant because the case scenario states that the wireless network was open and could be accessed by multiple people. Therefore, further evidence, particularly browser or cookie information, is required to associate the device with a specific individual.

Wireshark interface showing a packet capture of an HTTP POST request. The filter is `eth.addr == 00:17:f2:e2:c0:ce`. The selected packet is a POST request to `/secure/submit HTTP/1.1` from `192.168.15.4` to `192.168.1.106` on port 80. The packet details show the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
83593	15192.160911	192.168.15.4	208.185.127.40	TCP	64	35898 → 80 [ACK] Seq=299 Ack=387 Win=6
83594	15195.451766	192.168.15.4	192.168.1.106	TCP	82	36042 → 3283 [SYN] Seq=0 Win=65535 Len=0
83596	15196.388296	192.168.15.4	192.168.1.106	TCP	82	[TCP Retransmission] 36042 → 3283 [SYN] Seq=0 Win=65535 Len=0
83597	15197.118251	192.168.15.4	69.25.94.22	TCP	82	36044 → 80 [SYN] Seq=0 Win=64240 Len=0
83598	15197.205802	69.25.94.22	192.168.15.4	TCP	78	80 → 36044 [SYN, ACK] Seq=0 Ack=1 Win=0
83599	15197.210545	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=1 Ack=1 Win=64296
83600	15197.215799	192.168.15.4	69.22.167.247	TCP	70	35956 → 80 [FIN, ACK] Seq=365 Ack=1491
83601	15197.216422	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)
83602	15197.223434	69.22.167.247	192.168.15.4	TCP	70	80 → 35956 [FIN, ACK] Seq=1491980 Ack=365
83603	15197.311456	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [ACK] Seq=1 Ack=650 Win=649
83604	15197.373891	69.25.94.22	192.168.15.4	HTTP	359	HTTP/1.1 302 Moved Temporarily
83605	15197.373985	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [FIN, ACK] Seq=290 Ack=650
83606	15197.376297	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=290 Win=6
83607	15197.376686	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=291 Win=6
83608	15197.377312	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [FIN, ACK] Seq=650 Ack=291 Len=0

Packet details for the selected packet (83601):

- Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
- Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
- TCP, Src Port: 36044, Dst Port: 80
- Hypertext Transfer Protocol, Method: POST, URI: /secure/submit, Version: HTTP/1.1

No.	Time	Source	Destination	Protocol	Length	Info
83063	15157.378233	192.168.15.4	69.25.94.22	TCP	70	35994 → 80 [ACK] Seq=292 Ack=2737 Win=64296
83064	15157.379033	192.168.15.4	69.25.94.22	TCP	70	35994 → 80 [ACK] Seq=292 Ack=3428 Win=64296
83065	15157.379338	192.168.15.4	69.25.94.22	TCP	70	35994 → 80 [ACK] Seq=292 Ack=3429 Win=64296
83066	15157.405393	192.168.15.4	69.25.94.22	TCP	70	35994 → 80 [FIN, ACK] Seq=292 Ack=3429 Win=64296
83067	15157.412688	192.168.15.4	69.25.94.22	TCP	82	36002 → 80 [SYN] Seq=0 Win=64240 Len=0
83071	15157.434217	192.168.15.4	69.25.94.22	TCP	70	36000 → 80 [ACK] Seq=1 Ack=1 Win=64296
83072	15157.435074	192.168.15.4	69.25.94.22	HTTP	356	GET /images/body-bk.gif HTTP/1.1
83086	15157.502295	192.168.15.4	69.25.94.22	TCP	70	36002 → 80 [ACK] Seq=1 Ack=1 Win=64296
83087	15157.503801	192.168.15.4	69.25.94.22	HTTP	358	GET /images/bttn-send.gif HTTP/1.1
83091	15157.503929	192.168.15.4	69.25.94.22	TCP	70	36000 → 80 [ACK] Seq=287 Ack=2737 Win=64296
83093	15157.536361	192.168.15.4	69.25.94.22	TCP	70	36000 → 80 [ACK] Seq=287 Ack=4105 Win=64296
83099	15157.601265	192.168.15.4	69.25.94.22	TCP	70	36002 → 80 [ACK] Seq=289 Ack=1168 Win=64296
83100	15157.601607	192.168.15.4	69.25.94.22	TCP	70	36002 → 80 [ACK] Seq=289 Ack=1169 Win=64296
83101	15157.606558	192.168.15.4	69.25.94.22	TCP	70	36002 → 80 [FIN, ACK] Seq=289 Ack=1169 Win=64296
83103	15157.623974	192.168.15.4	69.25.94.22	TCP	70	36000 → 80 [ACK] Seq=287 Ack=5473 Win=64296

Wireshark interface showing a packet capture of an HTTP GET request. The filter is `eth.src == 00:17:f2:e2:c0:ce && ip.dst == 69.25.94.22`. The selected packet is a GET request to `/secure/success HTTP/1.1` from `192.168.15.4` to `192.168.1.106` on port 80. The packet details show the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
83175	15157.855200	192.168.15.4	69.25.94.22	TCP	70	36008 → 80 [FIN, ACK] Seq=292 Ack=825 Win=64296 Len=0
83597	15197.118251	192.168.15.4	69.25.94.22	TCP	82	36044 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=1
83599	15197.210545	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=1 Ack=1 Win=64296 Len=0 MSS=1460 WS=1
83601	15197.216422	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)
83606	15197.376297	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=290 Win=64007 Len=0 TSval=650
83607	15197.376686	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=291 Win=64007 Len=0 TSval=650
83608	15197.377312	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [FIN, ACK] Seq=650 Ack=291 Win=64296 Len=0 TSval=650
83609	15197.379749	192.168.15.4	69.25.94.22	TCP	82	36046 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=1
83613	15197.467914	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=1 Ack=1 Win=64296 Len=0 MSS=1460 WS=1
83614	15197.468887	192.168.15.4	69.25.94.22	HTTP	461	GET /secure/success HTTP/1.1
83620	15197.509811	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=392 Ack=2737 Win=62928 Len=0 TSval=650
83622	15197.574675	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=392 Ack=4105 Win=64296 Len=0 TSval=650
83624	15197.600935	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=392 Ack=5473 Win=62928 Len=0 TSval=650
83627	15197.664449	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=392 Ack=6841 Win=64296 Len=0 TSval=650
83629	15197.665819	192.168.15.4	69.25.94.22	TCP	70	36046 → 80 [ACK] Seq=392 Ack=8209 Win=62928 Len=0 TSval=650

Packet details for the selected packet (83620):

- Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
- Internet Protocol Version 4, Src: 192.168.15.4, Dst: 192.168.1.106
- TCP, Src Port: 36046, Dst Port: 80
- Hypertext Transfer Protocol, Method: GET, URI: /secure/success, Version: HTTP/1.1

The filter for Wireshark `eth.src == 00:17:f2:e2:c0:ce && ip.dst == 69.25.94.22` shows:

- Source MAC: 00:17:f2:e2:c0:ce
- Source IP: 192.168.15.4
- Destination IP: 69.25.94.22
- Relevant frame: 83601
- Protocol: HTTP
- Request: POST /secure/submit HTTP/1.1
- Displayed packets: 101

Most importantly, the multiple packets from 192.168.15.4 to 69.25.94.22, including HTTP traffic shows that the MAC address is associated with the client traffic communicating with the willselfdestruct.com server.

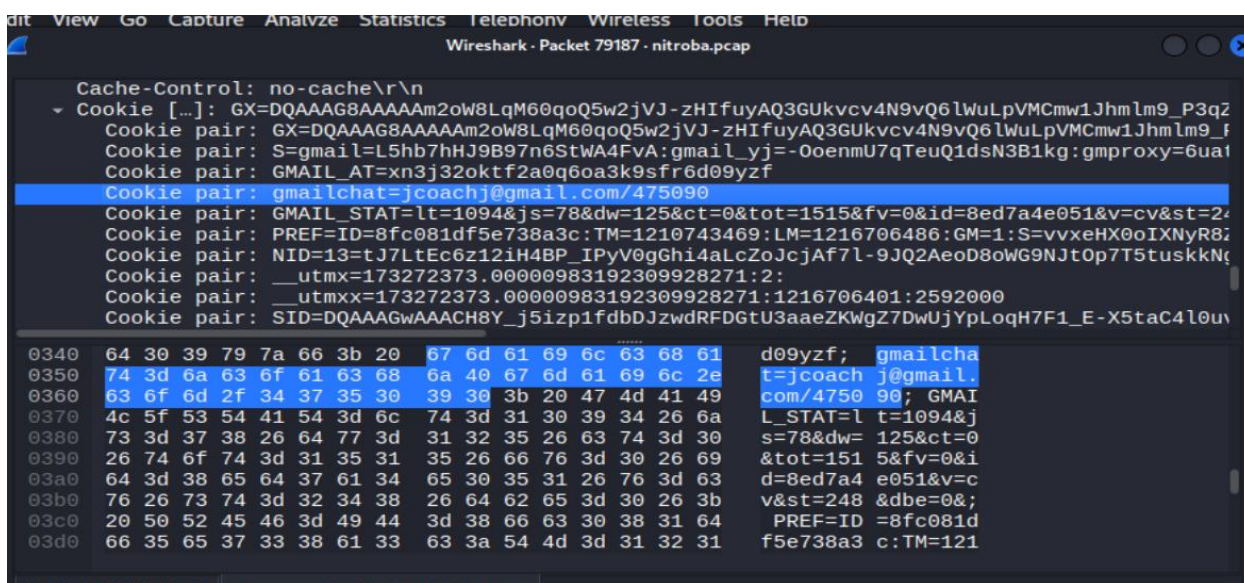
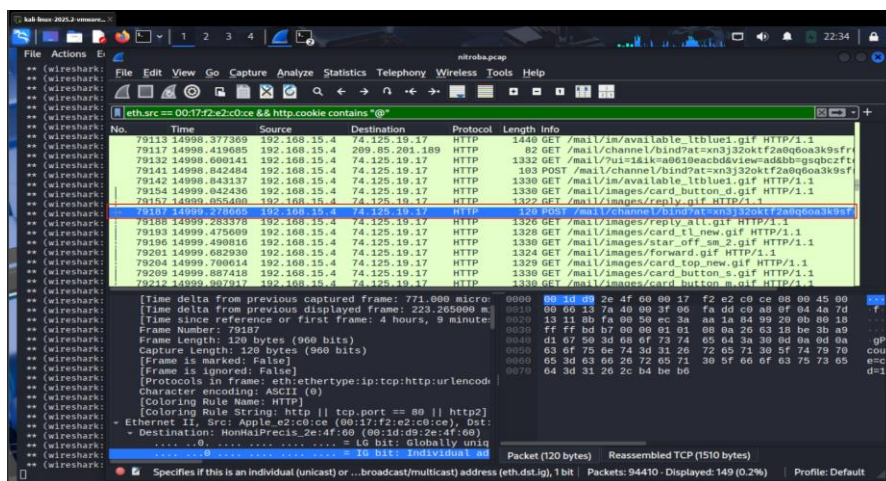
Evidence ID	Packet	Finding	Why it matters
E04	83601	Source MAC 00:17:f2:e2:c0:ce is associated with client IP 192.168.15.4, which generated the relevant HTTP traffic to 69.25.94.22.	Establishes the network interface/device associated with the captured harassment request.

The relevant request originated from the network interface identified by MAC address 00:17:f2:e2:c0:ce, using IP address 192.168.15.4.

Association of the Device With an Individual

Further analysis of the HTTP traffic associated with MAC address 00:17:f2:e2:c0:ce identified browser cookie evidence that provides an association between the device and a specific Gmail account. In Frame 79187, the HTTP Cookie field contains the cookie pair `gmailchat=jcoachj@gmail.com/475090`. The packet originates from client IP address 192.168.15.4 and is associated with the same MAC address, 00:17:f2:e2:c0:ce, identified in Question 4. The presence of `jcoachj@gmail.com` within the `gmailchat` cookie provides browser/session-level evidence linking the observed device to the Gmail account. This evidence strengthens the attribution from the network device to an identifiable account; however, the cookie alone does not conclusively establish that the account owner was

physically operating the device when the harassment request was made. The account association should therefore be considered together with the case scenario and student roster evidence before reaching an individual attribution.



- **Device MAC:** 00:17:f2:e2:c0:ce
- **Client IP:** 192.168.15.4
- **Frame:** 79187
- **Cookie:** gmailchat=jcoachj@gmail.com/475090
- **Account identified:** jcoachj@gmail.com

The evidence above establishes an association between the device/browser session and the Gmail account.

Evidence ID	Packet	Finding	Why it matters
E05	79187	HTTP cookie contains gmailchat=jcoachj@gmail.com/475090, while the traffic originates from MAC 00:17:f2:e2:c0:ce.	Provides browser/session evidence associating the device identified in Q4 with the Gmail account jcoachj@gmail.com.

Verification Against the Chemistry 109 Class Roster

The Gmail account identified through the HTTP cookie evidence was jcoachj@gmail.com. Frame 79187 contains the cookie pair gmailchat=jcoachj@gmail.com/475090, linking the observed browser session on MAC address 00:17:f2:e2:c0:ce to the Gmail account.



So who did it?

Chemistry 109 class list:

Teacher: Lily Tuckrige

Students:

Amy Smith
 Burt Greedom
 Tuck Gorge
 Ava Book
 Johnny Coach
 Jeremy Ledvkin
 Nancy Colburne
 Tamara Perkins
 Esther Pringle
 Asar Misrad
 Jenny Kant

The supplied Chemistry 109 class list includes a student named **Johnny Coach**, whose name corresponds to the jcoachj account identifier. Therefore, the individual associated with the identified Gmail account is listed as a student in Chemistry 109. This provides an additional link between the device identified in the network capture and a student who was enrolled in the class taught by Lily Tuckrige. However, the roster match should be

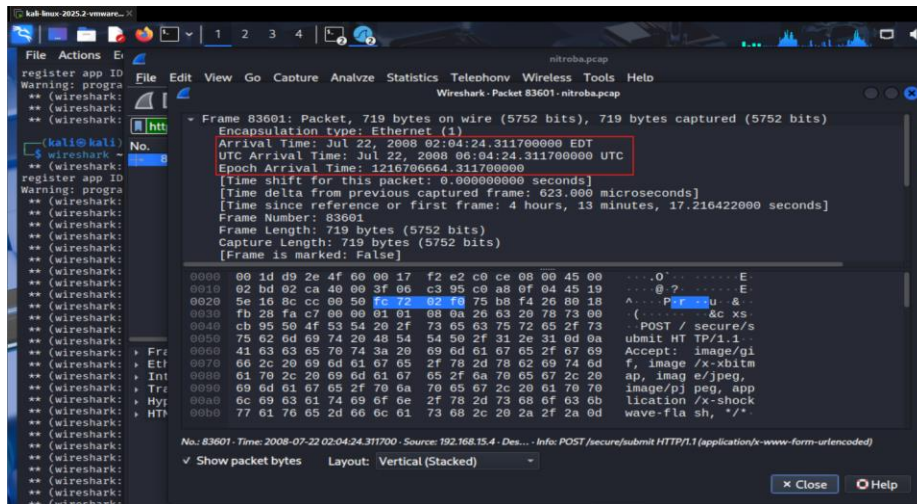
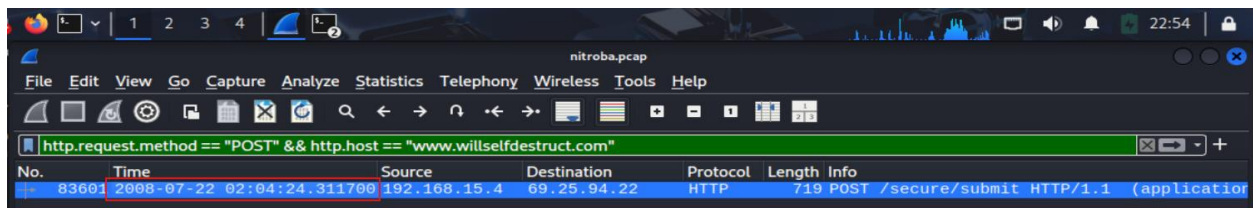
considered together with the network and browser evidence rather than treated as independent proof that Johnny Coach personally operated the device at the exact time of the harassment request.

Evidence ID	Evidence	Finding	Significance
E06	Chem 109 class list + Frame 79187	jcoachj@gmail.com corresponds to Johnny Coach, who appears on the Chemistry 109 roster.	Links the identified Gmail account to a student enrolled in Lily Tuckrige's class.

Activity Timeline

The relevant harassment activity occurred at 02:04:24.311700 EDT on July 22, 2008, according to the packet capture. Frame 83601, which contains the HTTP POST /secure/submit request to www.willselfdestruct.com, has an Arrival Time of Jul 22, 2008 02:04:24.311700 EDT and a corresponding UTC Arrival Time of Jul 22, 2008 06:04:24.311700 UTC. The packet's epoch timestamp is 1216706664.311700000. Wireshark also records the packet as occurring 4 hours, 13 minutes, and 17.216422 seconds after the first captured frame. The PCAP timestamp therefore places the relevant request shortly after midnight on July 22 in the capture's displayed Eastern Daylight Time. Although the case scenario describes the later harassment as occurring on Monday, July 21, the timestamp recorded in the PCAP is July 22, 2008, so the report uses the captured timestamp as the primary forensic time reference rather than changing it to match the scenario description. The key point is that Wireshark shows the packet as:

- Frame: 83601
- Arrival Time: July 22, 2008 02:04:24.311700 EDT
- UTC Arrival Time: July 22, 2008 06:04:24.311700 UTC
- Epoch: 1216706664.311700000
- Time since first frame: 4:13:17.216422



Evidence ID	Packet/item	Finding	Why it matters
E07	Frame 83601	Arrival Time: Jul 22, 2008 02:04:24.311700 EDT; UTC: Jul 22, 2008 06:04:24.311700 UTC	Establishes the recorded time of the HTTP harassment submission

Our evidence chain so far

Question	Evidence	Finding
Q2	Frame 82936	192.168.15.4 contacted www.willselfdestruct.com
Q3	Frame 83601	Actual harassment POST containing Lily's email/message
Q4	Frame 83601	Source MAC 00:17:f2:e2:c0:ce
Q5	Frame 79187	Same MAC/IP associated with jcoachj@gmail.com
Q6	Case slide, PDF p.14	Johnny Coach is on Chemistry 109 roster

Question	Evidence	Finding
Q7	Frame 83601	Submission at Jul 22, 2008 02:04:24.311700 EDT

Final Evidence Chain

Frame 83601

Actual harassment POST



192.168.15.4



MAC 00:17:f2:e2:c0:ce



Frame 79187

Same IP + MAC



gmailchat=jcoachj@gmail.com/475090



Johnny Coach



Chemistry 109 student

Forensic Contribution and Attribution Assessment

Based on what we've established:

1. Frame 83601 contains the harassment submission to lilytuckrige@yahoo.com.
2. The request originated from 192.168.15.4.
3. The Ethernet source MAC was 00:17:f2:e2:c0:ce.
4. The same MAC is associated with HTTP traffic containing the Gmail cookie: gmailchat=jcoachj@gmail.com/475090
5. jcoachj@gmail.com corresponds to Johnny Coach, who is on the Chemistry 109 class list.
6. The case scenario says the wireless network was open and accessible to multiple people.

7. Therefore, the evidence creates a strong association between the harassment traffic, a particular device, a Gmail account, and a Chem 109 student, but it does not independently prove that Johnny Coach physically operated the device at the exact moment the message was submitted.

The evidence supports a strong network and account-level association between the harassment message and Johnny Coach, a student listed in Chemistry 109. Frame 83601 contains the HTTP POST /secure/submit request to www.willselfdestruct.com, with the message addressed to lilytuckrige@yahoo.com. The request originated from client IP address 192.168.15.4 and Ethernet MAC address 00:17:f2:e2:c0:ce. Further traffic from the same MAC address contains the HTTP cookie `gmailchat=jcoachj@gmail.com/475090` in Frame 79187, providing browser/session evidence associating the device with the Gmail account jcoachj@gmail.com. The supplied Chemistry 109 class list identifies Johnny Coach as a student in Lily Tuckrige's class. Taken together, these findings provide a consistent evidentiary link from the harassment request to a specific network device, from that device to a Gmail account, and from the account identifier to a student enrolled in Chemistry 109.

However, the evidence does not conclusively establish that Johnny Coach personally operated the device when the harassment message was submitted. The case scenario states that the wireless router was open and passwordless and that multiple people could access the network. Therefore, the shared network environment creates an alternative explanation in which another person could have used the device or network connection. In addition, a browser cookie demonstrates an account/session association but does not by itself prove who was physically using the computer at the precise time of Frame 83601. The appropriate forensic conclusion is therefore that the evidence strongly supports the attribution of the relevant device and browser activity to an account associated with Johnny Coach, who was a Chemistry 109 student, but does not provide absolute proof of individual authorship of the harassment message. Further corroborating evidence, such as endpoint artifacts, user logon records, router/DHCP records, or additional account activity tied to the same timestamp, would be required to strengthen attribution to the individual beyond the network and browser evidence captured in this PCAP.

EVIDENCE REPOSITORY AND RECAP

ID	Evidence	Key Finding	Significance
E01	nitroba.pcap	SHA-256 verified against published hash	Establishes integrity of primary evidence
E02	PCAP metadata	94,410 packets; Ethernet; microsecond timestamps; 4h 22m 40s capture	Establishes characteristics of evidence examined
E03	Frame 83601	HTTP POST containing harassment message	Direct network-level evidence of submission
E04	Frame 83601 / MAC analysis	MAC 00:17:f2:e2:c0:ce associated with 192.168.15.4	Identifies network interface/device
E05	Frame 79187	gmailchat=jcoachj@gmail.com/475090	Associates browser session with Gmail account
E06	Chemistry 109 roster	Johnny Coach corresponds to identified account and is on roster	Links account to student enrolled in class
E07	Frame 83601 timestamp	Jul 22 2008 02:04:24.311700 EDT / 06:04:24.311700 UTC	Establishes recorded time of submission

LIMITATIONS

1. Open/shared wireless access: The scenario states that the wireless router was open and passwordless. Therefore, identifying the device and its IP/MAC address does not automatically identify the person who was using it.
2. **Cookie ≠ physical user:** The gmailchat=jcoachj@gmail.com/475090 cookie links the browser session to the Gmail account, but a cookie does not establish

with certainty who was physically sitting at the computer when the harassment request occurred.

OVERALL CONFIDENCE ASSESSMENT

What the PCAP supports strongly:

Harassment POST

↓

192.168.15.4

↓

MAC 00:17:f2:e2:c0:ce

↓

Gmail cookie: jcoachj@gmail.com

↓

Johnny Coach

↓

Chemistry 109 student

What the PCAP does NOT conclusively prove:

Johnny Coach was physically sitting at the computer and personally submitted the message.

Thus, the overall confidence in the findings is assessed at different levels because the available evidence supports some conclusions more strongly than others.

Moderate-to-High Confidence: Device, Account, and Student Association

There is moderate-to-high confidence that the network device identified by MAC address 00:17:f2:e2:c0:ce, using client IP address 192.168.15.4, was associated with the browser session belonging to the Gmail account jcoachj@gmail.com. This assessment is supported by the correlation of the network-layer and browser-layer evidence. Frame 83601 shows the device identified by MAC address 00:17:f2:e2:c0:ce using IP address 192.168.15.4 to submit the HTTP POST /secure/submit request containing the harassment message. Separately, Frame 79187 shows traffic from the same MAC address containing the HTTP cookie gmailchat=jcoachj@gmail.com/475090. The supplied Chemistry 109 class list contains Johnny Coach, whose name corresponds to

the identified account identifier. These findings provide a consistent chain of evidence connecting the relevant network activity to a particular device, a Gmail account, and a student enrolled in Chemistry 109.

Lower Confidence: Definitive Identification of the Physical User

There is lower confidence in concluding that Johnny Coach was the person physically operating the device at the exact time the harassment message was submitted. The PCAP provides network and browser/session evidence but does not directly identify the person who was physically using the computer at the time of Frame 83601. The case scenario also states that the wireless router was open and passwordless and that multiple people could access the network. Consequently, another individual could potentially have accessed the network or used the device associated with the captured traffic.

The Gmail cookie provides an association between the browser session and jcoachj@gmail.com, but the presence of an account identifier in a browser cookie does not independently establish who was physically operating the computer when the HTTP POST request occurred. Similarly, the correspondence between jcoachj@gmail.com and Johnny Coach, together with his presence on the Chemistry 109 roster, strengthens the attribution but does not constitute independent proof of physical presence or authorship.

Confidence Statement

Based on the evidence examined, the investigation has moderate-to-high confidence that the harassment request was generated through the identified network device and that the device's browser activity was associated with the Gmail account jcoachj@gmail.com, which corresponds to Johnny Coach, a Chemistry 109 student. However, there is lower confidence in making a definitive statement that Johnny Coach personally authored or submitted the harassment message, because the available evidence does not eliminate the possibility that another person accessed or operated the device.

Accordingly, the findings should be interpreted as a strong evidentiary association rather than conclusive proof of individual authorship. Additional evidence such as endpoint examination, operating-system user logon records, browser history, router or DHCP records, authentication records, or other activity directly tied to the relevant timestamp could provide stronger support for individual attribution.

CONCLUSION

The forensic examination established a consistent evidentiary chain connecting the harassment submission to a specific network device and subsequently to a Gmail account associated with a Chemistry 109 student. Frame 83601 captured the relevant HTTP POST request to www.willselfdestruct.com, originating from IP address 192.168.15.4 and MAC address 00:17:f2:e2:c0:ce. The request contained the harassment message addressed to lilytuckrige@yahoo.com. Frame 79187, originating from the same MAC address, contained the HTTP cookie `gmailchat=jcoachj@gmail.com/475090`, associating the observed browser session with the Gmail account jcoachj@gmail.com. The supplied Chemistry 109 roster identifies Johnny Coach as a student, providing an additional association with the class taught by Lily Tuckrige.

The evidence therefore supports a strong association between the captured harassment activity, the identified device, the Gmail account, and Johnny Coach as a Chemistry 109 student. Nevertheless, because the case scenario identifies the wireless network as open and accessible to multiple users, and because browser cookie evidence does not independently establish the physical identity of the person operating the device, the evidence does not conclusively prove that Johnny Coach personally authored or submitted the message. The defensible forensic conclusion is that the available PCAP provides substantial evidence supporting the device/account/student association, while definitive individual attribution would require additional corroborating evidence.